MAT 221 - CALCULUS I
Lesson 1 - Review of the Basic Formulas of Integration

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Example 1

Evaluate the following integrals

(a) $\int \left(4x^6 - \frac{3}{\sqrt{x}} + e^{-3x}\right) dx$ (b) $\int \left(5^x + \frac{4}{1+x^2} - \frac{7}{x}\right) dx$

(c) $\int \left(x^{\frac{1}{3}} + \frac{1}{\sqrt{1-x^2}} - \sin 2x\right) dx$

(d) $\int \left(5\sqrt{x} + \sec^2 5x - 5\cos 3x\right) dx$

Solution

(a) $\int \left(4x^6 - \frac{3}{\sqrt{x}} + e^{-3x}\right) dx = \frac{4x^7}{7} - 6\sqrt{x} - \frac{1}{3}e^{-3x} + C$

(b) $\int \left(5^x + \frac{4}{1+x^2} - \frac{7}{x}\right) dx = \frac{1}{\ln 5}5^x + 4\tan^{-1} x - 7\ln x + C$

(c) $\int \left(x^{\frac{1}{3}} + \frac{1}{\sqrt{1-x^2}} - \sin 2x\right) dx = \frac{3}{4}x^{\frac{4}{3}} + \sin^{-1} x + \frac{1}{2}\cos 2x + C$

(d) $\int \left(5\sqrt{x} + \sec^2 5x - 5\cos 3x\right) dx = \frac{10}{3}x^{\frac{3}{2}} + \frac{1}{5}\tan 5x - \frac{5}{3}\sin 3x + C$

Integration

- **Integration or anti-derivative** of a continuous function $f(x)$, is the reverse process of differentiation.
- **Indefinite Integration**
  Let F be an anti-derivative for f; then the indefinite integral of f is written

$$\int f(x)\,dx = F(x) + C$$

- **The Definite Integral**
  Let f be defined on $[a, b]$ with $a < b$. Then the definite integral of f over the interval $[a, b]$, is given by

$$\int_a^b f(x)\,dx = F(b) + F(a)$$

Review Of The Basic Formulas Of Integration

1. $\int u^n du = \frac{u^{n+1}}{n+1} + C, \quad n \neq -1;$
2. $\int \frac{1}{u} du = \ln|u| + C$
3. $\int a^u du = \frac{1}{\ln a} a^u + C$
4. $\int e^u du = e^u + C$
5. $\int \cos u du = \sin u + C;$
6. $\int \sin u du = -\cos u + C$
7. $\int sec^2 u du = \tan u + C;$
8. $\int \csc^2 u du = -\cot u + C$
9. $\int \sec u \tan u du = \sec u + C$
10. $\int \csc u \cot u du = -\csc u + C$
11. $\int \tan u du = -\ln|\cos u| + C;$
12. $\int \cot u du = \ln|\sin u| + C$
13. $\int \sec u du = \ln|\sec u + \tan u| + C;$
14. $\int \csc u du = -\ln|\csc u + \cot u| +$
15. $\int \frac{du}{\sqrt{1-u^2}} = \sin^{-1} u + C = -\cos^{-1} u + C$
16. $\int \frac{du}{1+u^2} = \tan^{-1} u + C = -\cot^{-1} u + C$
17. $\int \frac{du}{a^2+u^2} = \frac{1}{a} \tan^{-1} \frac{u}{a} + C;$
18. $\int \frac{du}{u\sqrt{u^2-1}} = \sec^{-1}|u| + C$

Example 2

Compute

(a) $\int_1^2 (3x^2 - x^5)dx.$ (b) $\int_1^4 \left(\frac{3}{\sqrt{s}} - 5\sqrt{s}\right) ds.$ (c) $\int_0^{\frac{\pi}{2}} \sin x dx.$

Solution

(a) $\int_1^2 (3x^2 - x^5)dx = \left(\frac{3x^5}{5} - \frac{x^6}{6}\right)_1^2 =$

$$\left[\frac{3(2)^5}{5} - \frac{2^6}{6}\right] - \left[\frac{3(1)^5}{5} - \frac{1^6}{6}\right] = \frac{81}{10}$$

(b) $\int_1^4 \left(\frac{3}{\sqrt{s}} - 5\sqrt{s}\right) ds = \left(6\sqrt{s} - \frac{10}{3}s^{\frac{3}{2}}\right)_1^4 = -\frac{52}{3}.$

(c) $\int_0^{\frac{\pi}{2}} \sin x dx = -\cos x|_0^{\frac{\pi}{2}} = -\cos\frac{\pi}{2} - (-\cos 0) = 1.$

Integration By Substitution

Theorem 1

If $u = g(x)$ is a differentiable function of x and $du = g'(x)dx$ is its differential, then if $r \neq -1$,

$$\int [g(x)]^r g'(x)dx = \int u^r du.$$

Example 3

Calculate

(a) $\int \sqrt{3+x}\, dx$ (b) $\int_0^1 (x^3-1)^{\frac{11}{5}} 3x^2\, dx$ (c) $\int x\sqrt[3]{1+x^2}\, dx$

(d) $\int \cos 3x\, dx$ (e) $\int \frac{\sin x}{1-\cos x}\, dx$ (f) $\int \frac{x-1}{x^2-2x+3}\, dx$

Solution

(a) Let $u = g(x) = 3 + x$. Then $du = dx$, and

$$\int \sqrt{3+x}\, dx = \int \sqrt{u}\, du = \frac{2}{3}u^{\frac{3}{2}} = \frac{2}{3}(3+x)^{\frac{3}{2}} + C.$$

(b) - (f) solution in class.